

Italian aristocrat Alessandro Volta was a meticulous investigator whose experiments revealed that electricity can be generated chemically, as well as by friction. Chief among his many scientific contributions was the invention of the Voltaic pile, the world's first battery, which laid the path for the coming age of electric power.

Born in Como, Italy, Alessandro Volta was the only one of seven siblings not to become a priest, having met scientist and physician Giuliano Gattoni. At the age of 23, he wrote his first paper on electricity, and at 29, he became professor of physics at Como's Royal School.

Volta was quick to make scientific breakthroughs. In 1775, he improved the electrophorus, a resin disk that could be rubbed with cat fur to make a static charge—the only kind of electricity known at that time. But it was Luigi Galvani's 1786 finding that a dissected frog's legs could be made to twitch by electricity that sparked him into action. Rejecting this idea of "animal electricity," Volta theorized that contact between different metals created a chemical reaction, which produced electricity. He then tested different combinations of metals by placing them on his tongue, resulting in a mild electric shock. This led to his 1799 discovery that stacking disks of zinc, copper, and brine-soaked paper in sequence produced an electric current, which increased as more disks were added. His "Voltaic pile" was the first battery and the prime means of creating electricity until the discovery of electromagnetic generation in 1829.

MILESTONES

GAS DISCOVERY

Captures methane while collecting marshland air in 1776. Ignites the gas with a spark inside a glass jar.

ACADEMIC POST

In 1779, becomes professor of experimental physics at the University of Pavia, where he stays for 40 years.

MAKES ELECTRICITY

Generates electric current from a circuit of copper, zinc, and brine-soaked paper in 1791.

FIRST BATTERY

Creates the "Voltaic pile" in 1799, the first apparatus for generating and storing electricity.



Volta held demonstrations that transfixed a public eager for electrical thrills. In 1801, French emperor Napoleon invited him to Paris to promote his scientific experiments.